Assignment 1 - Product Requirement Document - YouTube Answer Mode - Saurav Dasgupta C52

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PRODUCT REQUIREMENTS DOCUMENT (PRD)

Answer Mode

Search that understands questions, not just videos.

Product: YouTube

Feature: Answer Mode (MVP)

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Executive Summary

Answer Mode is an optional search experience designed for YouTube users who come to the platform with a specific question rather than a general intent to browse videos. Instead of returning only videos, Answer Mode identifies and surfaces the most relevant sections within long-form videos that are likely to answer the user's query.

Research conducted during Phase 2 showed that knowledge seekers spend a significant amount of time determining whether a video contains the information they need and then manually searching within the video to locate the relevant moment. This creates unnecessary friction, particularly for educational videos, lectures and podcasts.

Answer Mode addresses this problem by changing the unit of retrieval from an entire video to the most relevant moments within it. Users can immediately jump to the section most likely to answer their question while retaining the ability to continue watching the original video if they find it valuable.

The objective is not to replace YouTube Search or long-form content, but to help knowledge seekers reach relevant information faster while preserving engagement with creators.

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1. Problem & Evidence

Problem Statement (P0)

Knowledge seekers searching for answers on YouTube struggle to determine whether a video contains the information they need. Even when the answer exists within a video, they often spend considerable time manually navigating long timelines before reaching the relevant section, making the process slow, frustrating and unpredictable.

This problem is particularly visible for educational videos, interviews, lectures and podcasts where answers are embedded within long-form discussions rather than clearly segmented chapters.

Evidence from User Research

The problem statement is based on findings from the user interviews and survey conducted during Phase 2.

The strongest observations were:

- **21 of 26 participants** said they typically opened multiple videos before finding one that answered their question.
- **18 of 26 participants** reported manually scrubbing through long timelines because they did not know where the relevant explanation was located.
- **19 of 26 participants** said they had abandoned useful long-form videos simply because finding the answer required too much effort.
- **Only 5 of 26 participants** regularly relied on timestamps or chapters, with several noting that podcasts and discussion videos rarely contained meaningful navigation markers.

These findings indicate that the primary friction is not the lack of educational content on YouTube. Rather, it is the effort required to discover where the answer exists within that content.

Why This Matters

YouTube has become one of the largest repositories of educational knowledge on the internet.

However, its search experience continues to retrieve videos rather than answers.

For users who arrive with a learning objective, the platform currently asks them to complete two separate tasks:

1. Determine whether a video contains the answer.
2. Find the moment where that answer appears.

Both tasks require manual effort.

Reducing this effort has the potential to improve the overall search experience without fundamentally changing how YouTube works for entertainment-focused users.

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2. User & Context

Primary User

The primary audience for Answer Mode is **knowledge seekers**.

These users come to YouTube with a clearly defined question and are looking for an explanation rather than entertainment.

Typical examples include:

- Students
- Software engineers
- Working professionals
- Researchers
- UPSC / CAT / GRE aspirants
- Hobby learners
- People troubleshooting technical problems

Unlike casual viewers, these users usually know what they want to learn before opening YouTube.

Context of Use

The feature is designed for informational searches such as:

- How does a fusion reactor work?
- Explain binary search trees.
- How do I configure Kubernetes networking?

- Why does inflation occur?
- How does photosynthesis work?

The user is typically looking for one specific explanation rather than an entire video.

Where the Journey Breaks Today

The breakdown occurs immediately after search results are shown.

Today's experience looks like this:

Search

↓

Open Video

↓

Watch Intro

↓

Skip Forward

↓

Skip Again

↓

Open Another Video

↓

Repeat

↓

Eventually Find Answer

↓

Leave

The user spends a disproportionate amount of time navigating videos rather than learning.

This behavior appeared consistently across the interviews and was one of the strongest themes emerging from the research.

Opportunity

Instead of asking users to discover the right video and then discover the correct moment within that video, YouTube can directly surface the most relevant sections that answer the user's query.

The user's effort shifts from searching for information to evaluating explanations.

This reduces navigation while preserving the value of long-form educational content.

3. Goals

The objective of Answer Mode is to improve the experience for knowledge seekers without changing YouTube's existing search experience for other users.

The MVP focuses on four goals.

Goal 1

Reduce the time required for users to locate relevant answers within long-form videos.

Goal 2

Increase user confidence that a returned result contains the information they are looking for before they start watching.

Goal 3

Reduce unnecessary video hopping and manual timeline scrubbing during informational searches.

Goal 4

Preserve engagement with long-form creator content by allowing users to seamlessly continue watching the full video after discovering the answer.

4. Non-Goals

To keep the MVP focused, several ideas have been intentionally excluded from this release.

Answer Mode is **not** intended to:

- Replace the existing YouTube Search experience.
- Automatically enable itself for every search.
- Generate answers independently of creator content.
- Replace educational videos with AI-generated responses.
- Personalize search results based on individual learning history.
- Recommend the "best creator."
- Summarize entire videos.
- Build a conversational AI assistant.

These ideas may be explored in future iterations but are outside the scope of the MVP.

5. Product Principles

The following principles guided every design decision made during the development of Answer Mode.

Principle 1 — Answers before Videos

For informational searches, users care about reaching the answer as quickly as possible.

Answer Mode should reduce the effort required to locate that answer without reducing the value of the original video.

Principle 2 — Explain Every Recommendation

Users should never have to guess why a particular section was recommended.

Every surfaced section should clearly communicate why it matched the search query using simple, human-readable explanations rather than technical confidence scores.

Principle 3 — Preserve Creator Value

Answer Mode should encourage users to continue engaging with the original content.

Finding the answer should become the beginning of the viewing journey rather than the end.

This is why every answer section includes a clear **Watch Full Video** option.

Principle 4 — Keep YouTube Familiar

Entertainment and informational searches represent different user intents.

Rather than replacing YouTube Search entirely, Answer Mode should remain an optional experience that users can enable when they are looking for answers.

Principle 5 — Reduce Navigation, Not Curiosity

The objective is not to shorten videos.

The objective is to eliminate unnecessary navigation so users can spend more time learning and less time searching.

6. Solution Overview

Proposed Solution

The proposed solution is **Answer Mode**, an optional search experience designed specifically for knowledge-seeking behavior on YouTube.

When Answer Mode is turned **off**, YouTube Search behaves exactly as it does today.

When Answer Mode is turned **on**, the platform changes the unit of retrieval from **videos** to **answer-bearing moments within videos**.

Instead of asking users to discover whether a video contains the answer and then manually locate the relevant section, Answer Mode identifies the most relevant moments across YouTube's video library and presents them directly within the search results.

The objective is not to replace videos with AI-generated answers. Every answer remains grounded in creator-generated content and acts as an entry point into the original video.

Prototype

Live Prototype

Walkthrough Video

Insert Walkthrough Video Link

Experience Overview

The Answer Mode experience consists of four stages:

1. Enable Answer Mode
2. Search using a natural language question
3. Review answer-centric search results
4. Jump directly to the most relevant moment within a video

Each stage has been designed to remove a specific friction identified during user research.

End-to-End User Flow

Step 1 — Enable Answer Mode

The experience begins on the existing YouTube Home screen.

Users who are looking for information can enable **Answer Mode** using a toggle placed beside the search bar.

This decision was intentional.

Research showed that YouTube serves multiple user intents. While informational searches benefit from answer-centric retrieval, entertainment and discovery-oriented browsing do not. Making Answer Mode optional preserves the familiarity of the existing search experience while introducing a specialized mode for users with a learning objective.

With the toggle disabled, YouTube Search remains unchanged.

Step 2 — Search

Once Answer Mode is enabled, users search exactly as they do today.

Example searches include:

- How do fusion reactors work?
- Explain Kubernetes networking.
- What causes inflation?
- How does binary search work?

No new syntax or query language is required.

Users simply ask questions naturally.

Step 3 — Answer-Centric Search Results

Instead of returning only videos, Answer Mode returns the **single most relevant section from each video** that best answers the user's query.

Each search result becomes an **Answer Card**.

An Answer Card contains:

- Video title
- Creator name
- Thumbnail
- Timestamp of the recommended section
- Short summary of that section
- Relevance label
- Explanation of why the section matched
- Watch This Moment
- Watch Full Video
- More Relevant Sections (if applicable)

The intention is to help users evaluate whether a result is worth opening before they start watching.

Relevance Labels

Instead of displaying technical confidence scores such as **98% Match** or **76% Match**, Answer Mode uses simple qualitative labels.

Possible labels include:

- **Best Match**
- **Highly Relevant**
- **Relevant**

During solution exploration, numerical scores were considered but ultimately rejected.

While they appeared precise, they provided little practical value to users and could create a false sense of accuracy. Interviews suggested that users care far more about understanding whether a section is worth watching than the exact numerical confidence assigned by a ranking model.

The chosen labels communicate confidence in a simpler and more human way.

Why This Section Matched

Every Answer Card explains why the section was selected.

For example:

Why this section matched

✓ Explains the process of nuclear fusion

✓ Covers the working principle of Tokamak reactors

✓ Discusses plasma confinement

This feature serves two purposes.

First, it increases user confidence by making the recommendation transparent rather than asking users to trust the ranking algorithm blindly.

Second, it allows users to quickly determine whether the section covers the aspect of the topic they are interested in before opening the video.

Research showed that users wanted to know *why* a recommendation was relevant, not simply that it was relevant.

Multiple Relevant Sections

Many educational videos answer the same question at several points.

Instead of displaying every matching section immediately, Answer Mode initially shows only the strongest matching interval for each video.

If additional answer-bearing sections exist, users see a button such as:

2 More Relevant Sections

Selecting this expands the Answer Card and reveals:

- Additional timestamps
- Short summaries
- Relevance labels
- Why each section matched

This decision balances completeness with simplicity.

Showing every matching interval upfront would make the results visually overwhelming, especially for long podcasts or lectures. Progressive disclosure allows interested users to explore deeper without increasing cognitive load for everyone else.

Watch This Moment

Users who select **Watch This Moment** begin playback directly from the beginning of the recommended section.

This removes one of the largest sources of friction identified during research: manually scrubbing through long videos in search of a relevant explanation.

The feature reduces the number of interactions required to reach the answer while preserving the continuity of the original video.

Video Player Experience

The video player has been enhanced to support answer-centric navigation while remaining recognizably YouTube.

Three additions are introduced.

Timeline Highlights

Every answer-bearing section identified for the current search query is highlighted on the video's timeline.

These highlights allow users to understand where relevant explanations occur throughout the video at a glance.

Rather than repeatedly scrubbing through the timeline, users can move directly between meaningful sections.

Relevant Moments Panel

A new panel appears alongside the video.

Each relevant section is displayed together with:

- Timestamp
- Short summary
- Relevance label
- Explanation of why the section matched

Selecting any item immediately jumps playback to the beginning of that section.

This provides users with a structured overview of the video's answer content without requiring them to manually explore the timeline.

Watch Full Video

Answer Mode is designed to help users reach the answer—not to discourage long-form viewing.

At any point, users can choose **Watch Full Video** and continue watching the original content naturally.

This supports one of the product's core principles: preserving creator value.

The answer becomes an entry point into the full video rather than a replacement for it.

Section Feedback

After watching a suggested section, users can provide lightweight feedback using a simple thumbs-up or thumbs-down interaction.

Unlike liking or disliking an entire video, this feedback answers a much more specific question:

Did this section answer your search query?

This distinction is important.

A user may enjoy a video overall while still feeling that a particular section was not relevant to their search.

Over time, these feedback signals can be incorporated into YouTube's ranking system to improve the quality of future answer recommendations for similar queries.

Keeping the interaction limited to a single click minimizes effort while still generating valuable relevance data.

Design Decisions

Several design decisions were made deliberately during the solution exploration process.

Decision	Rationale
Answer Mode is optional	Different users come to YouTube with different intents. Entertainment searches should continue using the existing experience.
Qualitative labels replace confidence scores	Users make decisions based on perceived usefulness rather than numerical precision.
Only the strongest section is shown initially	Prevents overwhelming users with too much information while keeping additional matches accessible.
Every recommendation explains why it matched	Increases transparency and user confidence in the ranking system.
Watch Full Video remains prominent	Encourages continued engagement with creator content rather than replacing it.
Section-level feedback instead of video-level feedback	Captures whether the surfaced answer actually solved the user's query, creating a more useful signal for future ranking improvements.

MVP Scope

The first release intentionally focuses on solving one problem well.

The MVP includes:

- Answer Mode toggle
- Answer-centric search results
- Best Match / Highly Relevant / Relevant labels
- Why this section matched
- Multiple relevant sections
- Watch This Moment
- Watch Full Video
- Timeline highlights
- Relevant Moments panel
- Section-level thumbs up/down feedback

The following ideas were intentionally excluded from the MVP and will be evaluated in future iterations:

- Next Best Answer navigation across videos
- Refine Your Answer using topic filters
- Watch History support for answer sections

- Cross-video AI-generated summaries
- Personalized answer ranking

Restricting the MVP to the core answer-discovery experience allows the team to validate the primary hypothesis before investing in more advanced capabilities.

7. Functional Requirements

The MVP focuses on helping knowledge seekers reach relevant answers within long-form videos with the least possible effort. The following functional requirements define the expected behavior of Answer Mode.

FR-1: Answer Mode Toggle

Description

Users should be able to enable or disable Answer Mode directly from the YouTube Home page before performing a search.

When disabled, YouTube Search should continue to function exactly as it does today.

Acceptance Criteria

- Answer Mode is OFF by default.
- Users can enable or disable it with a single tap.
- Search behavior changes only when the toggle is enabled.
- User preference is remembered for future sessions until changed.

FR-2: Answer-Centric Search Results

Description

When Answer Mode is enabled, search results should display the highest-ranked answer section from each matching long-form video instead of presenting only complete videos.

Each Answer Card should include:

- Video title
- Creator name
- Thumbnail
- Recommended timestamp
- Short summary of the section

- Relevance label
- Why this section matched
- Watch This Moment
- Watch Full Video
- More Relevant Sections (if applicable)

Acceptance Criteria

- Every result displays one primary answer section.
- Results remain ordered according to overall relevance.
- Every Answer Card clearly distinguishes the recommended section from the rest of the video.

FR-3: Relevance Labels

Description

Each answer section should display a qualitative relevance label.

Possible labels include:

- Best Match
- Highly Relevant
- Relevant

These labels replace numerical confidence scores.

Acceptance Criteria

- Every returned section has exactly one relevance label.
- Labels remain consistent across Search Results and Video Player.
- Numerical confidence percentages are not displayed.

FR-4: Why This Section Matched

Description

Every answer section should explain why it was selected.

Example:

Why this section matched

- Includes explanation of nuclear fusion
- Explains Tokamak reactor design

- Covers plasma confinement

Acceptance Criteria

- Every Answer Card displays at least one explanation.
- Explanations use simple language rather than technical model outputs.
- Explanations remain visible inside the Video Player.

FR-5: Multiple Relevant Sections

Description

If multiple sections from the same video answer the query, only the strongest section should appear initially.

Users can expand the Answer Card to reveal additional relevant sections.

Acceptance Criteria

- Initial search results display one section per video.
- Additional sections are accessible through **More Relevant Sections**.
- Expanded sections include summaries, timestamps and relevance labels.

FR-6: Watch This Moment

Description

Selecting **Watch This Moment** should begin playback directly from the beginning of the selected answer section.

Acceptance Criteria

- Playback starts at the selected timestamp.
- The selected section is highlighted.
- Users can immediately begin watching without manual navigation.

FR-7: Video Player Enhancements

Description

The Video Player should support answer-focused navigation.

Enhancements include:

- Highlighted answer sections on the timeline.

- Relevant Moments panel.
- Section summaries.
- Relevance labels.
- Why this section matched.
- Direct navigation between answer sections.

Acceptance Criteria

- Every matched section is highlighted on the timeline.
- Relevant Moments panel displays all matched sections.
- Selecting a section immediately updates playback.
- Timeline highlighting remains synchronized with playback.

FR-8: Watch Full Video

Description

Users should be able to continue watching the complete video from any answer section.

Acceptance Criteria

- Watch Full Video remains available throughout playback.
- Selecting it resumes normal viewing without leaving the video.
- Viewing history behaves like standard YouTube playback.

FR-9: Section-Level Feedback

Description

Users should be able to indicate whether a recommended section answered their question using simple feedback controls.

Options:

👍 Answered my question

👎 Didn't answer my question

This feedback is intended to improve future ranking quality.

Acceptance Criteria

- Feedback requires a single interaction.
- Feedback is recorded only once per viewed section.

- Users can change their feedback if submitted accidentally.

8. User Stories

User Story 1

As a knowledge seeker,

I want to know whether a video contains the answer to my question before opening it,

so that I do not waste time watching irrelevant videos.

Acceptance Criteria

- Search results display why each section matched.
- Relevant sections are visible before playback begins.

User Story 2

As a learner,

I want to jump directly to the part of the video that answers my question,

so that I don't have to manually scrub through long timelines.

Acceptance Criteria

- Watch This Moment starts playback from the selected section.
- Timeline navigation is no longer required to locate the answer.

User Story 3

As a learner,

I want to understand why a particular section was recommended,

so that I can quickly decide whether it is worth watching.

Acceptance Criteria

- Every section displays one or more "Why this section matched" explanations.
- Explanations are easy to understand.

User Story 4

As a learner,

I want to explore additional answer sections from the same video,

so that I can compare different explanations without searching again.

Acceptance Criteria

- More Relevant Sections expands the Answer Card.
- Additional sections display summaries and timestamps.
- Selecting one begins playback from that section.

User Story 5

As a viewer,

I want to continue watching the complete video after finding my answer,

so that I can explore the topic in greater depth if I choose.

Acceptance Criteria

- Watch Full Video remains accessible throughout playback.
- Users can transition seamlessly from answer-focused viewing to normal viewing.

User Story 6

As a user,

I want to tell YouTube whether a recommended section actually answered my question,

so that future recommendations become more accurate.

Acceptance Criteria

- Users can provide thumbs-up or thumbs-down feedback.
- Feedback is optional.
- The interaction does not interrupt playback.

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9. Edge Cases

The MVP should gracefully handle situations where Answer Mode cannot confidently identify answer-bearing sections.

Scenario	Expected Behaviour
No matching answer sections found	Fall back to standard YouTube search results with a message explaining that no specific answer could be identified.

Scenario	Expected Behaviour
Video has no transcript	Show the video in standard search results without Answer Mode enhancements.
Transcript quality is poor	Suppress answer extraction and return the standard video result.
Very short videos	Treat the entire video as a single answer section if appropriate.
Livestream recordings	Process only if searchable transcripts are available.
Multiple identical answer sections	Display the strongest instance and suppress duplicates.
Deleted or private videos	Exclude them from Answer Mode results.
Age-restricted content	Follow existing YouTube access rules before surfacing answer sections.
User disables Answer Mode mid-session	Immediately revert to the standard YouTube search experience.

10. Empty States

Good empty states help users understand why Answer Mode cannot provide enhanced results without making the product feel broken.

Empty State 1

No Answer Sections Found

Message

> *We couldn't find a specific section that answers your question. Here are the standard YouTube search results instead.*

Primary Action

Browse Videos

Empty State 2

No Additional Relevant Sections

When only one relevant section exists, the **More Relevant Sections** button is hidden.

No placeholder or empty container is displayed.

Empty State 3

Transcript Not Available

Message

> *Answer Mode isn't available for this video because searchable transcripts are unavailable.*

The video remains accessible through the standard YouTube experience.

Empty State 4

No Feedback Submitted

If the user has not rated a section, display the standard thumbs-up and thumbs-down controls.

Once feedback is submitted, update the selected state while allowing the user to change their choice later.

Empty State 5

No Search Query

If Answer Mode is enabled but the search box is empty, the interface behaves exactly like the existing YouTube Home screen.

Answer Mode becomes active only after a search is performed.

Requirements Summary

The MVP is considered functionally complete when users can:

- Enable Answer Mode.
- Search using natural language questions.
- Discover answer-centric search results.
- Understand why each section was recommended.
- Explore additional relevant sections within a video.
- Jump directly to relevant moments.
- Navigate between answer sections inside the Video Player.
- Continue watching the complete video.
- Provide simple section-level feedback to improve future recommendations.

11. Success Metrics

The success of Answer Mode will be measured by evaluating whether it helps knowledge seekers find answers faster without reducing engagement with creator content.

The metrics below focus on the specific user segment identified during Phase 2 rather than the overall YouTube user base.

Key Metric (North Star)

Answer Discovery Success Rate

Definition

Percentage of Answer Mode searches where users begin watching a recommended answer section without returning to the search results within the same session.

Why this metric?

The core objective of Answer Mode is to help users reach the answer they are looking for more efficiently.

If users consistently find relevant answer sections and continue watching instead of immediately returning to search for another video, it indicates that the recommendation successfully addressed their query.

This metric directly measures whether the product solves the P0 identified during research.

Secondary Metric 1

Time to First Relevant Answer

Definition

Median time between submitting a search query and beginning playback of a relevant answer section.

Why this metric?

One of the biggest pain points identified during user interviews was the time spent opening multiple videos and manually scrubbing through timelines.

A reduction in this metric would indicate that users are locating relevant information more efficiently.

Secondary Metric 2

Answer Section Engagement Rate

Definition

Percentage of users who continue watching beyond the recommended answer section or choose **Watch Full Video**.

Why this metric?

The goal of Answer Mode is not simply to deliver an answer but to act as a better entry point into long-form content.

A healthy engagement rate would indicate that users continue exploring creator content after finding the information they need.

Guardrail Metric

Long-form Watch Time

Definition

Average watch time of long-form educational videos viewed through Answer Mode compared with standard YouTube Search.

Why this metric?

A potential concern is that surfacing answer sections could reduce viewing time by encouraging users to watch only short portions of videos.

Monitoring long-form watch time ensures that improving answer discovery does not unintentionally reduce creator engagement.

If watch time drops significantly, the experience may need adjustment—for example, by making transitions to the full video more prominent.

Success Metrics Summary

Metric	Success Indicator	Why it Matters
North Star	Higher Answer Discovery Success Rate	Measures whether users are finding relevant answers more effectively.
Secondary	Lower Time to First Relevant Answer	Confirms that Answer Mode reduces search effort.
Secondary	Higher Answer Section Engagement Rate	Shows whether users continue exploring after finding an answer.
Guardrail	Stable or improved Long-form Watch Time	Ensures creator engagement is not negatively impacted.

12. Rollout Strategy

Introducing Answer Mode changes one of YouTube's most frequently used experiences. To minimize risk, the feature should be rolled out gradually.

Phase 1 – Internal Dogfooding

Audience:

- YouTube Search team
- Product Managers
- UX Researchers
- Designers
- Engineers

Objectives:

- Validate ranking quality.
- Identify obvious usability issues.
- Verify recommendation explanations.
- Collect qualitative feedback before exposing the feature externally.

Success Criteria:

- Stable search performance.
- No critical usability issues.
- Positive internal feedback.

Phase 2 – Limited Beta

Audience:

A small percentage of logged-in users who frequently perform educational or informational searches.

Objectives:

- Validate user adoption.
- Measure answer discovery success.
- Evaluate feedback quality.
- Compare behavior with existing search.

Success Criteria:

- Improved Answer Discovery Success Rate.
- Reduced Time to First Relevant Answer.
- No significant drop in long-form watch time.

Phase 3 – Public Launch

Audience:

All eligible YouTube users.

Characteristics:

- Answer Mode remains optional.
- Standard YouTube Search continues to be the default experience.
- Ongoing monitoring of key metrics.
- Continuous improvement of ranking quality using section-level feedback.

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13. Risks

The success of Answer Mode depends on several assumptions. The following risks have been identified for the MVP.

Risk	What Failure Looks Like	Likelihood	Impact	Early Signal	Mitigation
Answer sections are not relevant enough	Users quickly abandon recommended sections and return to search results.	Medium	High	Low Answer Discovery Success Rate and high bounce rate.	Begin with educational content that has high-quality transcripts. Continuously improve ranking using section-level feedback.
Users do not understand Answer Mode	Users ignore the toggle or continue using standard search.	Medium	Medium	Low Answer Mode adoption despite relevant search traffic.	Add lightweight onboarding during first use and keep the toggle visible near the search bar.
Creator watch time decreases	Users watch only answer sections and leave without engaging further.	Low	High	Decline in average watch time for Answer Mode sessions.	Keep **Watch Full Video** prominent and encourage seamless continuation into the full video.

Risk	What Failure Looks Like	Likelihood	Impact	Early Signal	Mitigation
Poor transcript quality affects recommendations	Incorrect or incomplete answer sections are surfaced.	Medium	Medium	High rate of negative section feedback.	Exclude videos with unreliable transcripts from Answer Mode until transcript quality improves.
Users misuse feedback controls	Feedback reflects video preference rather than answer relevance.	Low	Medium	Inconsistent feedback patterns across similar queries.	Frame feedback explicitly as "Did this section answer your question?" and combine it with behavioral signals before influencing ranking.

14. Open Questions

The MVP intentionally leaves several questions unanswered. These will require further research and cross-functional discussion before future iterations.

1. Which searches should trigger Answer Mode?

Should it be enabled only for informational searches, or should users be able to use it for any search query?

This decision would require input from the Search and UX teams.

2. How should relevance labels be assigned?

What criteria determine whether a section is labelled **Best Match**, **Highly Relevant**, or **Relevant**?

The exact thresholds would need to be defined collaboratively by Search Ranking and Machine Learning teams.

3. How much context should be included around the answer?

Starting exactly at the answer may reduce understanding, while adding too much context may increase viewing time unnecessarily.

Additional user research would be needed to determine the optimal balance.

4. How should conflicting user feedback influence ranking?

If one user marks a section as helpful while another does not, how should the ranking system interpret these signals?

The ranking model would likely need to combine explicit feedback with behavioural metrics such as watch time, rewatches, and subsequent search activity.

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5. Should creators have visibility into Answer Mode analytics?

Would it be useful for creators to understand which sections of their videos are most frequently surfaced as answers?

This has implications for both creator tools and recommendation quality and would require input from the Creator Product team.

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15. Future Roadmap

The MVP focuses on validating the core hypothesis: that surfacing answer-bearing moments improves the search experience for knowledge seekers.

Several ideas discussed during product exploration have been intentionally deferred to future phases.

Phase 2 Enhancements

- **Next Best Answer:** Allow users to jump directly to the best matching section in the next recommended video without returning to search results.
- **Refine Your Answer:** Enable users to filter results using topic chips derived from the "Why this section matched" explanations (e.g., *Tokamak Reactor*, *Plasma Confinement*).
- **Answer Section History:** Extend Watch History to track viewed answer sections in addition to full videos, ensuring continuity across sessions.

Phase 3 Enhancements

- **Cross-video AI Summary:** Generate a concise synthesis of insights from the highest-ranked answer sections while clearly linking back to the original creator content.
- **Personalized Ranking:** Incorporate user learning preferences and previous interactions to improve answer relevance over time.
- **Learning Collections:** Allow users to save and organize answer sections into thematic collections for future reference.

Closing Note

Answer Mode is built around a simple idea: when users come to YouTube with a question, they should spend less time searching for the right moment and more time learning from it.

Rather than replacing videos, the feature makes them easier to navigate by surfacing the most relevant sections, explaining why they were selected, and preserving a clear path back to the full creator content. The MVP deliberately focuses on solving this one problem well, leaving more advanced discovery and personalization features for future iterations after the core experience has been validated.

Appendix

A. Design Decisions Considered

During solution exploration, several approaches were considered before arriving at the final design.

Option	Decision	Reason
Return full videos only	Rejected	Does not solve the core problem of locating the answer within long-form content.
Display numerical relevance scores (e.g., 92% Match)	Rejected	Users care more about understanding why a section is relevant than seeing confidence percentages, which can create a false sense of precision.
Automatically enable Answer Mode for all searches	Rejected	Users come to YouTube with different intents. Entertainment and discovery searches should continue using the existing search experience.
Display all matching sections upfront	Rejected	Creates visual clutter, especially for podcasts and lectures. Showing the strongest section first keeps results easier to scan while allowing users to explore additional matches if needed.
Generate standalone AI answers	Rejected	Shifts value away from creator content and fundamentally changes YouTube's role. The objective is to improve discovery, not replace videos.

Out of Scope (Future Iterations)

The following ideas were intentionally excluded from the MVP to keep the initial release focused on validating the core hypothesis.

Phase 2

- Next Best Answer navigation across videos
- Refine Your Answer using topic-based filters

- Watch History support for answer sections

Phase 3

- Cross-video AI-generated summaries
- Personalized answer ranking
- Multi-video answer synthesis
- Creator analytics for Answer Mode
- Saved answer collections

Dependencies

Successful implementation of Answer Mode depends on collaboration across several teams.

Team	Responsibility
Search Ranking	Ranking answer-bearing sections
Machine Learning	Section retrieval, relevance scoring and explanation generation
Transcript Platform	High-quality transcript availability
UX Design	Search results and Video Player experience
YouTube Player	Timeline highlights and section navigation
Data Science	Metrics definition and experiment analysis
Trust & Safety	Handling misinformation and low-quality recommendations

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Launch Checklist

Before public rollout, the following conditions should be met:

- Answer extraction quality validated across major educational categories.
- Relevance labels tested with users.
- "Why this section matched" explanations reviewed for clarity.
- Watch This Moment playback tested across platforms.
- Timeline highlighting works correctly for videos with multiple answer sections.
- Feedback collection pipeline operational.
- Success metrics instrumented and verified.
- Accessibility review completed.
- Performance impact on Search remains within acceptable limits.

Product Vision

Answer Mode reimagines YouTube Search for users who arrive with a specific question rather than a desire to browse.

Instead of asking users to first discover the right video and then search within it for the relevant moment, the experience surfaces the most useful sections directly while preserving the richness of long-form creator content. Users spend less time navigating and more time learning, while creators continue to benefit from deeper engagement with their videos.

Document References

Prototype:

https://claude.ai/public/artifacts/72535428-7de1-4141-bb98-f633c92ec44d

Prototype Walkthrough Video:

https://drive.google.com/file/d/1Q0cpYMpVCsXNFHsIw1qjgwx_w0tO8e2g/view?usp=drive_link

Phase 1 Research:

> Assignment_1_-_Phase_1_-_Saurav_Dasgupta_C52.pdf

Phase 2 Research:

> Assignment_1-_Phase_2_-_Saurav_Dasgupta_C52.pdf

Revision History

Version	Date	Description
1.0	29/08/2026	Initial PRD for Answer Mode MVP